

Xujiang Tang

Theoretical Mathematics Researcher | Machine Learning Theory Enthusiast

Undergraduate Student in Mathematics and Applied Mathematics, Yangtze University | Visiting Student, Great Bay University

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RESEARCH INTERESTS

- Algebraic topology and equivariant homotopy theory: fixed point methods, geometric fixed point tomography, Bokstedt periodicity, and stable homotopy-theoretic structures.
- Finite group theory: Quillen's p-subgroup conjecture, subgroup complexes, centralizer structures, and FRUL-WOS sphere methods.
- Formal proof: Lean, Mathlib, and the formalization of advanced mathematics in algebra and topology.
- Machine learning theory: generalization, learnability, out-of-distribution prediction, and Koopman/operator-theoretic methods for long-sequence data.

EDUCATION

Expected Jun. 2027	Yangtze University College of Arts and Sciences B.Sc. in Mathematics and Applied Mathematics. Selected coursework: mathematical analysis, linear algebra, probability and statistics, differential equations, computational modeling, optimization theory, numerical computation, data structures, Python, R, and MATLAB. Selected marks: Optimization Theory 97/100; Numerical Computation Methods 96/100.	Jingzhou, China
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VISITING POSITION AND TRAINING

2025-present	Visiting Student, Great Bay University Research in theoretical mathematics and machine learning theory.	
2026	Zhejiang University Lean Formalization Summer School Formal proof, Lean, Mathlib, and computer-assisted theorem proving.	
Winter training	Neural Networks and Python Implementation Short winter training with Prof. Mark Vogelsberger. Completed the course and received an A grade.	Grade: A

RESEARCH EXPERIENCE

2025-present	Finite Group Theory and Quillen's p-Subgroup Conjecture Centralizer energy, FRUL-WOS spheres, and subgroup-complex methods related to Quillen's p-subgroup conjecture.
2025-present	Algebraic Topology and Equivariant Homotopy Theory Geometric fixed point tomography and spoke Bokstedt periodicity.
2026-present	Formal Proof and Lean Lean formalization, proof engineering, and the long-term formalization of advanced mathematical arguments.
2024-present	Theoretical Machine Learning and Optimization Learnability, generalization guarantees, out-of-distribution prediction, and optimization-theoretic aspects of learning algorithms.
2023-present	Stochastic Processes, Dynamical Systems, and Quantitative Finance Stochastic-process models, rough volatility, MCMC-based simulation, Koopman/operator-theoretic dynamics, and robust financial decision problems.
2023-2025	Epidemic Prevention and Population Mobility Modeling, Jingzhou Mathematical modeling and statistical surveys related to epidemic prevention and population mobility in Jingzhou.
2025-2026 H1	Spatial Omics Data Processing Spatial omics data processing with Dayu Hu, Northeastern University.
2025-2026 H2	Epilepsy and Long-Sequence Data Research Epilepsy-related long-sequence modeling in Prof. Huaguang Gu's group, focusing on Koopman operator methods.

PUBLICATIONS AND MANUSCRIPTS

Published and accepted papers

1. Q. Li and X. Tang. "HKG EduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation." IEEE ICMEIM 2025, pp. 1-5. EI-indexed.
2. X. Tang and Q. Li. "Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning." 5th International Conference on Internet, Education and Information Technology (IEIT 2025), Hangzhou, China, 2025. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
3. Q. Li and X. Tang. "Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions." AIMS Mathematics. Accepted, 2025.

MANUSCRIPTS

1. X. Tang. "Centralizer Energy, FRUL-WOS Spheres, and Quillen's p-Subgroup Conjecture." Manuscript.
2. X. Tang. "Geometric Fixed Point Tomography and Spoke Bokstedt Periodicity." Manuscript.
3. X. Tang and C. Fan. "Out-of-Distribution Generalization Error Bounds for Successful Prediction of Deep Autoregressive Algorithms." Manuscript.
4. X. Tang. "Quartic Difficulty: Assessing the Learnability of Unsupervised Learning Algorithms." Manuscript.

COLLABORATORS AND MENTORS

- William Redman, Johns Hopkins University.
- Ziyue Qiao, Great Bay University.

SOFTWARE COPYRIGHTS

- Software copyright in financial systems.
- Software copyright in unmanned swarm systems.

HONORS, COMPETITIONS, AND ACTIVITIES

- Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge.
- Regional Gold Medal, WorldQuant International Quant Championship.
- Third Prize, UCAS Graduate AI Forum.
- Mathematical Modeling Competition, Hua Zhong Cup.
- National College Students Statistical Modeling Competition.
- Multiple national second prizes in mathematical modeling competitions.
- Reviewer, ICLR 2026 Workshop.

PROFESSIONAL EXPERIENCE

Jun.-Aug. 2024	Peking University - Open-Source LLM Group Contributed to LLM deployment and retrieval-augmented generation systems.
Jul.-Sep. 2024	Deloitte Professional experience in consulting workflows, teamwork, and communication.
Jul.-Sep. 2024	Guolian Securities Applied statistical methods to securities research and quantitative workflows.

SKILLS

- Mathematics: algebraic topology, finite group theory, homotopy theory, representation theory, optimization, numerical analysis, stochastic processes, differential equations, causal inference, statistical modeling.
- Formal proof and programming: Lean, Mathlib, Python, PyTorch, NumPy, Pandas, MATLAB, R, C++, LaTeX, Git/GitHub, Linux.
- Languages: Mandarin Chinese, English.